

Khine Zar Thwe

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PROFESSIONAL SUMMARY

Software engineer with experience developing and maintaining enterprise Java/Jakarta EE applications, including security and estate management systems deployed across Singapore. Proven track record of delivering production-grade mobile applications, integrating hardware security APIs, and collaborating with stakeholders to evaluate and integrate security product features. Skilled in Java, Spring Boot, REST API design, cloud technologies, and full software development lifecycle.

EXPERIENCE

Junior AI Developer – Reachfield IT Solutions Apr 2025 – Apr 2026 | Singapore

- Developed and deployed YOLO-based crowd-counting and object detection system achieving 85%+ accuracy across 30+ concurrent video streams with real-time intrusion detection.
- Built two production React Native apps (Work Order Management, Form Management) shipped to Google Play and TestFlight, demonstrating effective time management by deploying across 200+ active field staff on K10 and F2 ruggedised devices with full UAT coordination.
- Owned end-to-end Quality Assurance (QA) across production releases: conducted comprehensive UAT and bug triage, managed severity classification, wrote detailed bug reports and software documentation, and coordinated resolution across backend and mobile teams, ensuring clear technical writing for cross-functional stakeholders.
- Led Java-based application development by migrating Hibernate 5 → 6 and upgrading JDK 11 → 17 on the SGEMS2 platform, rewriting 50+ DAO/Service classes for Jakarta EE compliance, resolving transaction deadlocks to ensure system integrity, and performing security testing while coordinating UAT for 30+ critical bugs.
- Integrated hardware APIs: Face++ facial verification, Bluetooth thermal printers, QR/barcode scanners, GPS/EXIF tracking; implemented multi-flavour Android build configurations for device-specific requirements.
- Developed FastAPI Python microservices for real-time emotion detection (7-class custom CNN model) and crowd density analysis; deployed containerised inference server on Render with async request handling.
- Designed software architecture and implemented a robust REST API layer for SGEMS2, optimising SQL queries on 50 K+ records and training 20+ field staff on deployment and device configuration.
- Leveraged stakeholder management and technical collaboration to evaluate and integrate security product features, identifying potential vulnerabilities and implementing data protection improvements across the platform.
- Demonstrated advanced problem-solving and logical reasoning by conducting a root-cause analysis on CNN inference failures, leading to a 15% improvement in model accuracy through strategic hyperparameter tuning and architectural trade-off evaluations.

EDUCATION

Singapore Polytechnic – Singapore Apr 2026

Diploma in Information Technology (Software & Applications)

Elective: Certificate in Data & Analytics (2024)

CERTIFICATIONS

AWS Certified AI Practitioner (AIF-C01) – Amazon Web Services Jul 2026

AWS Cloud Practitioner (CLF-C02) Training Complete – Amazon Web Services Jun 2026

Software Testing/QA – LinkedIn Learning Jun 2026

AI Ethics & Governance (Associate Level) – Singapore Computer Society

S Pass Eligibility Letter – Singapore Polytechnic

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, C#, SQL, HTML/CSS

Frameworks & Libraries: React, React Native, Jakarta EE, Spring Boot, Node.js/Express, FastAPI, Hibernate, TensorFlow, OpenCV, MediaPipe, YOLO, CNN models, ASP.NET Core, .NET Core, EF Core

Web Technologies: Vite, TailwindCSS, REST APIs

Cloud & DevOps: AWS (Lambda, S3, Bedrock, API), Docker, Git, Linux, CI/CD

Databases: MySQL, PostgreSQL, Supabase, SQL Server

Mobile & Device Integration: Android Studio, Xcode, iOS/Android, Bluetooth, QR scanners, Face++ API, K10/F2 devices, GPS/EXIF

Testing & QA: UAT coordination, Bug severity classification, Bug report writing, Test case design, Regression testing, Manual testing, Postman (API testing)

FEATURED PROJECTS

Portfolio AI Chatbot — AWS Serverless (Bedrock + DynamoDB)

- Built a knowledge-grounded AI chatbot on the portfolio site using a vanilla-JS widget, API Gateway, and Python Lambda, with Amazon Bedrock (Claude Haiku 4.5) generating responses grounded in a system prompt of personal bio, skills, and projects.
- Implemented self-resetting serverless rate limiting in DynamoDB via atomic update_item with ConditionExpression keyed by year-month, capping usage at 300 messages/month with zero cron jobs or scheduled infrastructure.
- Engineered input validation (1000-char limit), conversation-history trimming to the last 6 turns, and graceful quota-exceeded fallback messaging for production-grade UX.

GlucoSG—Full-Stack Nutrition Tracking App

- Full-stack nutrition tracking app for diabetic population built with React/Vite frontend (HTML, CSS, JavaScript), Node/Express backend, and Supabase (PostgreSQL, realtime auth). Developed for a Singapore Polytechnic hackathon and won 1st place.

SGEMS2 Mobile Apps—React Native (Shipped)

- Two production React Native apps (Work Order Management, Form Management) shipped to Google Play and TestFlight. Deployed across 200+ field staff on rugged K10/F2 devices with full UAT and 30+ critical bugs resolved.

Code-to-Cloud Architecture Visualizer

- AWS-native application that converts pasted code into architecture diagrams with cost estimates using a React frontend, Python Lambda functions, API Gateway, and Claude Bedrock integration.

EmotionVision—Real-Time Emotion Detection

- Custom TensorFlow 7-class CNN model for emotion classification combined with OpenCV video pipeline and FastAPI microservices, containerised with Docker for real-time emotion and crowd density analysis.

SGEMS2 Enterprise Platform

- Java/Jakarta EE security and estate management system deployed across Singapore. Implemented DAO/Service layer, REST APIs, relational database schema design, JSP templating, and Hibernate ORM. Participated in comprehensive UAT cycle.

Banking QA Automation Framework

- Enterprise-grade end-to-end QA automation framework for banking applications built on ParaBank. Features dynamic test data selection, Page Object Model architecture, and automated testing on every commit via GitHub Actions with live Allure dashboard.

Hearth Coffee Co. — Custom Shopify Theme

2026-08-01

- Built a fully custom Shopify theme on Dawn (Online Store 2.0) with a from-scratch custom section, delivering theme-editor-level customisation in Liquid, JSON, and JavaScript.
- Created a custom “Find your roast” quiz section with no stock equivalent — posts scores to a self-built Node.js/Express API and hydrates results with live Shopify product data, with no Shopify app or Storefront API token required.
- Integrated a self-built service at runtime to demonstrate e-commerce skills beyond a themed storefront, covering custom sections, theme customisation, and API integration.

BookStock — E-Commerce Order & Inventory Management System

- Designed and built a layered (Clean Architecture) REST API in ASP.NET Core handling product catalog, order processing, mocked payments, and stock management, with JWT-based authentication
- Implemented atomic order transactions (stock validation, deduction, and payment creation) using SQL Server transactions with rollback handling to prevent overselling
- Wrote SQL Server stored procedures and views for reporting (monthly sales, top-selling products, low-stock alerts), reducing report query logic outside the application layer
- Deployed to Azure App Service with Azure SQL Database
- Built as a Singapore Polytechnic coursework project, demonstrating full-stack .NET development, database design, and cloud deployment from requirements to delivery.

LANGUAGES

English (Professional), Burmese (Native)